1. In object-oriented programming, create a class named Quadratic that finds the roots of a quadratic equation $ax^2 + bx + c = 0$ when the coefficients a, b, and c are given. Use the discriminant to determine whether the equation has two roots, one root, or no real roots.

File application.cpp

```
#include "quadratic.h"
int main ()
{
    Quadratic quad1 (2, 3, 1);
    quad1.print ();
    cout << endl;

    Quadratic quad3 (1, 4, 4);
    quad3.print();
    cout << endl;

    Quadratic quad4 (2 , 2, 4);
    quad4.print();
    return 0;
}
```

Output:

2x^2 + 3x + 1
Root 1: -0.5 Root 2: -1

1x^2 + 4x + 4
Root 1: -2 Root 2: -2

2x^2 + 2x + 4
The equation has no roots.

2. Using Zeller's congruence, design a class Zeller with data members for day, month, and year to find the corresponding weekday (Saturday through Friday).

```
weekday = (day + 26 * (month + 1) / 10 +
                                year + year / 4 – year /100 + year /400 ) % 7
```

File application.cpp

```
#include "zeller.h"
int main ()
{
    // Instantiation and printing weekday for first object
    Zeller z1 (3, 29, 2016);
    z1.print ();
            // Instantiation and printing weekday for second object
    Zeller z2 (5, 24, 2015);
    z2.print ();
            // Instantiation and printing weekday for third object
    Zeller z3 (8, 7, 2017);
    z3.print ();
    return 0;
}
```

Output:

3/29/2016 is Tuesday
5/24/2015 is Sunday
8/7/2017 is Monday

3. Design a Complex class representing complex numbers x + iy. Include member functions to add, subtract, multiply, and divide complex numbers, and print the results.

File application.cpp

```
#include "complex.h"
int main ()
{
    // Instantiation of four complex numbers
    Complex complex1 (2.0, 3.0);
    Complex complex2 (3.0, 4.0);
    Complex complex3 (5.0, 4.0);
    Complex complex4 (6.0, 5.0);
    // Instantiation a complex number to be used in operation
    Complex temp (2.0, 1.0);
    // Applying operations on the four complex numbers
    complex1.addTo (temp);
    complex2.subtractFrom (temp);
    complex3.multiplyBy (temp);
    complex4.divideBy (temp);
    // Printing the result of operations
    cout << "complex1 + temp   = ";
    complex1.print ();
    cout << "complex2 - temp   = ";
    complex2.print ();
    cout << "complex3 * temp   = ";
    complex3.print ();
    cout << "complex4 / temp   = ";
    complex4.print ();
    return 0;
}
```

Output:

```
complex1 + temp   = (+4.00) + i(+4.00)
complex2 - temp   = (+1.00) + i(+3.00)
complex3 * temp   = (+6.00) + i(+13.00)
complex4 / temp   = (+3.40) + i(+0.80)
```